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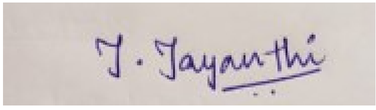

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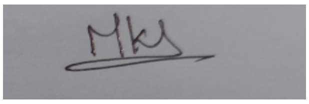

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Abstract

Recently the yoga has gained popularity around the world for its immense benefits. Since many people are adopting and integrating yoga into their daily routine , it is important to make sure that the poses are done perfectly to enjoy it's benefits. In te same way, performing yoga in a bad posture can also lead to various muscle issues and ligament tears Therefore posture is hold great importance when it comes to yoga. Therefore AI yoga trainers can help users keep a good posture by capturing data from the input video feed such as the angles between the joints and compares it with the ideal pose to make sure the users are on the right track.

An AI yoga trainer 's main functionality provide personalized yoga guidance and instruction. It analyzes a user's body movements and posture through the camera, and then suggests correcting yoga poses and sequences based on the individual's needs and abilities. This technology enables individuals to practice yoga at home with the guidance of a virtual trainer, making yoga more accessible and convenient for people of all levels and backgrounds.

Keywords - artificial intelligence, yoga, pre-trained model,

Introduction

The history of AI yoga trainers can be traced back to the development of artificial intelligence and its applications in various domains, including health and wellness. The concept of using AI to assist in yoga practice has gained popularity in recent years as technology has advanced and its potential benefits in the field of wellness have been recognized. The earliest forms of AI yoga trainers were basic applications that provided instructional videos or written instructions for different yoga poses. These early versions lacked sophisticated AI capabilities and were primarily focused on providing static information without any personalized feedback or adaptive features. They were limited in their ability to provide real-time guidance or customized instruction to meet the specific needs of individual practitioners.

As computer vision and motion tracking technologies evolved, AI yoga trainers began to incorporate these features to provide more interactive and personalized experiences. Computer vision allowed the AI system to analyze the practitioner's movements and provide feedback on their form and alignment, while motion tracking enabled the system to track the practitioner's movements in real-time and adjust the instructions accordingly. These advancements made AI yoga trainers more dynamic and capable of providing personalized guidance to practitioners. With the advent of machine learning, AI yoga trainers became more intelligent and capable of adapting to the needs and preferences of individual practitioners. Machine learning algorithms allowed the AI system to analyze data from multiple sources, such as the practitioner's body measurements, movement patterns, and feedback history, to provide personalized instructions and feedback. This enabled AI yoga trainers to provide tailored guidance based on the practitioner's skill level, fitness level, and goals.

The field of AI yoga trainers is still evolving, with ongoing research and development aimed at further improving their capabilities. As AI continues to advance, we can expect to see more sophisticated and intelligent AI yoga trainers that can provide even more personalized and adaptive guidance to practitioners. The integration of biometric sensors, wearable devices, and other health monitoring technologies may also enable AI yoga trainers to provide insights into the practitioner's physical well-being, stress levels, and other factors that may impact their yoga practice. Additionally, advancements in augmented reality (AR) and mixed reality (MR) may further enhance the immersive and interactive experiences offered by AI yoga trainers.

In the context of an AI yoga trainer, the ‘mediapipe.pose’ module can be utilized to estimate the pose of a practitioner during a yoga practice session.

The ‘mediapipe.pose’ module is designed to detect and track the 2D key points of a human body, such as the joints and landmarks, from a video or image input. These key points can then be used to infer the pose of the person, including the position and orientation of different body parts, such as the head, neck, shoulders, elbows, wrists, hips, knees, and ankles (as shown in figure 1).

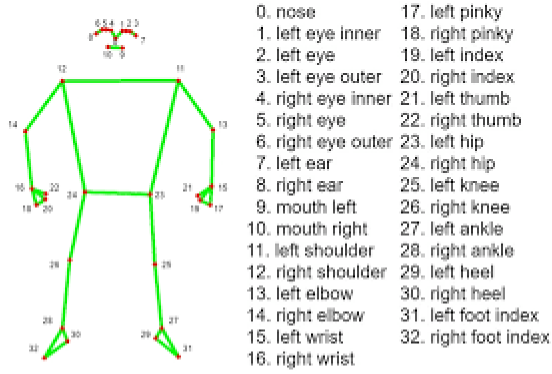


Fig 1. Mediapipe.pose landmarks

Literature Survey

A recent research paper [1], introduces 'YogGuru', which is an AI/ML-based yoga app that works not only by giving points to its users, guiding them in their daily yoga exercise, providing optimized suggestions and food recommendations but also integrates gaming for multiple players, and utilizes the various python libraries for detection and classification of human pose. It also makes use of Google's services to remind users for their daily yoga practice as well as guide them in their daily practice. The proposed system, YogGuru, offers many benefits to people who love to do yoga. YogGuru is an AI-based yoga app that offers optimized coaching, training step by step, gaming feature for multiple players, reminders, and a voice-guided trainer to ensure correct posture and prevent injuries. Some of the cons of YogGuru include potential inaccuracies in AI detection, high training time for model as well as limited dataset. The model developed by us has a very less execution time as it doesn't depend on the limited number of yoga postures (found in a dataset) that have been trained using various machine learning models. It provides an environment for the user to choose any yoga posture (not limited to the yoga postures dataset) and compare his posture to the actual yoga posture selected by him, by comparing the angles of the joints.

Another article [2], proposes a three-dimensional deep learning architecture based on CNN for recognizing yoga poses in real-world environments, obtaining high accuracy and showing the capability of using deep learning techniques in recognizing human activities, with possible applications in various field of life, especially, healthcare, fitness, and entertainment. The deep learning model that has been proposed in this paper, has various advantages such as high accuracy in recognizing different yoga poses, uninterrupted use of video as input, and multiple yoga postures, making it a valuable tool for healthcare, fitness, and entertainment industries. The cons of the deep learning model that has been proposed can be given as having a restricted number of recognized yoga postures, dependency on video as input, hardware requirements, and the absence of optimized response, but its correctness and flexibility still make it a valuable tool for monitoring yoga practice in real-world environments. Our model is capable of providing personal feedbacks, such as, telling the user which part of the body needs to be moved in order to achieve the correct position.

'Fitness Freaks,' a fitness tracker created using artificial intelligence techniques, is introduced by Pardeshi et al. [3] and addresses the problems of finding time for the gym, paying for a personal trainer, and the constraints of the global pandemic. It tracks gym exercise repetitions and detects incorrect body posture during yoga. The Fitness Freaks app has the advantages of allowing users to track their gym exercise repetitions, identify incorrect body posture while practising yoga, prevent injuries and pains, and get around the difficulties of finding time for the gym, paying for a personal trainer, and dealing with the restrictions of the global pandemic. This app includes drawbacks such as errors in the human posture estimation algorithm and restrictions on the kinds of workouts it can track. Our model is capable of tracking any type of yoga posture since it gives the user, the freedom to practice the yoga posture of his choice by uploading the picture of whichever pose he wants to try. It works by analyzing the angles made the joints in the picture and comparing it with the angles made by the user in real-time. It also provides personalized feedbacks.

One of the methods for accurately detecting and classifying yoga poses using machine learning techniques, specifically KNN, SVM, Naïve Bayes, and logistic regression, with an accuracy of 96% for the KNN model, using a dataset of 12 sun salutation yoga poses collected through a normal webcam and an estimation of pose algorithm, has been shown in another study [4]. This study's benefits include the use of pose identification methods to precisely categorise sun salutation yoga postures utilising machine learning models and real-time skeleton sketching for posture recognition. The method's drawbacks include a small dataset, a reliance on a camera, a constrained application, a high level of complexity, and a lack of comparison with other machine learning models or methodologies. In our model, we have incorporated the feature that allows the user to upload the image of the yoga pose that he wants to try, therefore not making the app restricted to only few types of yoga postures.

Ashraf et al. [5] have described how a deep learning-based model for recognising yoga poses, which uses spatial and depth features extracted from images and achieves higher accuracy (accuracy of 94.91% with 95.61% precision) than state-of-the-art image classification models, works and the benefits of depth-wise separable convolution. The suggested model is particularly well suited for at-home exercise settings since it provides greater performance, faster computation, requires less datasets, effectively uses various types of layers, can extract spatial and depth characteristics, and requires less data. The small dataset that could lead to overfitting, the unclear generalizability to other yoga poses, the high hardware requirements, the lack of interpretability, and the accuracy being influenced by human error in labelling or defining yoga poses are some of the possible drawbacks of the proposed model and approach. Since, in our work, we have introduced the feature of allowing the user to upload the image of the yoga pose that he wants to try, our model is not subjected to the risk of overfitting.

In this paper,'e-YogaGuru' is proposed which analyzes the body movements while performing yoga and provides feedback and instructions for matching the correct yoga pose if needed. This system takes dataset of 1750 video sequences(7 yogasana performed by 25 practitioners) as base to calculate the correctness of the posture done by the people.This system is only trained for 21 postures.It achieved an accuracy of 98.29% in identifying the Yogasana.The main advantage of the proposed system is the correctness of the posture is calculated by not only front view but also from side and back view using Kinect 1.0 sensor.This sensor is used for the dataset creation.The disadvantage of this model is it can automatically recognise only 21 postures of yogasana.

The ai-yoga trainer which we proposed will give a new feature to overcome the disadvantage of the above model.The new feature allows users to upload an image of their correct yogasana posture, which is then utilized to determine the accuracy of their yogasana posture through computation.Based on that feedback and instructions are provided to match the correct yoga pose.So this system does not depend on the dataset but on users input to calculate the correctness of the yoga posture [6].

In this paper,a system which automatically recognise yoga asanas in real time works based on deep learning algorithms is proposed.Initial task of this process is data collection which is done using RGB webcam.yoga poses of 15 individuals are collected and stored as dataset.open pose is used to calculate the joints and this data is given as input to Convolutional Neural Networks(CNN) to extract features.These features are given to LSTM to automatically recognise the yoga pose done by the users.The system achieved an accuracy of 98.92% when the system was tested on 12 persons(five males and seven females).
The main advantage of the proposed system is it can automatically recognise yoga pose not only in real time but in recorded videos.

The disadvantage of the system is it can only recognise six yoga asanas which are Bhujangasana, Padmasana, Shavasana, Tadasana, Trikonasana, and Vrikshasana.

Our proposed system which is called "ai-yoga trainer" doesn't only compute the correctness of the yoga pose but also give feedback and instructions to match the correct pose.By uploading the correct yoga pose beforehand, one can utilize the system to calculate the accuracy of any yoga posture [7].
In this paper,a system which is based on machine learning models like Logistic Regression,Random Forest,SVM is proposed to classify the yoga poses.The angles generated by the TF pose estimation algorithm is used to detect the yoga pose.The accuracy of 99.04% is achieved using Random forest classifier.
By creating a skeleton using the joint points, this system gains an advantage in accurately categorizing yoga poses. The system's ability to classify poses is improved by this feature.

The disadvantage is this system can only categorize the 10 yoga poses.
Our proposed system overcomes this disadvantage by helping the users to do any correct yoga pose and additionally it gives audio instructions for the users to do the correct pose [8].

In this work,the concept of fine-grained hierarchical pose classification and yoga-82, a dataset containing all 82 complex poses, is proposed.The dataset hierarchy consists of three levels.74% to 79.35% of accuracy is achieved when denseNet-201 algorithm is applied over the dataset to predict the labels of yoga poses.

The main advantage of this work is even the complex yoga poses can be predicted and they are categorized into three levels.The main disadvantage of this work is all the similar and new poses are categorized into the third level hierarchy of the dataset.Due to this accuracy of denseNet-201 algorithm is greatly affected.

Our proposed system does not compute the correctness of the yoga pose based on the dataset of yoga poses but it works on real time data.(based on the input provided by the user through webcam).The additional advantage is it gives feedback and audio instructions to perform correct yoga posture[9].
In this work,an IoT-based privacy-preserving yoga posture recognition system which works using deep convolutional neural networks (DCNN) and a low-resolution infrared sensor based wireless sensor network (WSN) is proposed.

The main advantage of this system is it will preserve the privacy of the users and the components used for detecting yoga poses are low-cost,unobtrusive,portable and easy to install.

The disadvantage of this model is that it can only detect 26 yoga poses in a certain environment as they are using the sensors with low resolution.
"Ai-yoga trainer" does not need any additional components for detecting yoga poses and it can help users to do correct yoga poses by giving feedback and audio instructions [10].

Proposed work

Our aim is to develop an AI Yoga Trainer which can taken real-time video input for yoga pose done by the user and can compare it with the reference video of yoga pose that has been uploaded. In order to achieve this, we have applied this pre-trained model ‘mediapipe’ to get real-time comparison of body angles. The CNN used in the mediapipe.pose module is trained on large datasets containing annotated human pose data, allowing it to learn to recognize the key points and connections between body joints that define human poses. During inference, the trained CNN takes input images or video frames and applies convolutional, pooling, and fully connected layers to extract relevant features and make predictions about the locations of body joints in the input data.

Input have been taken using ‘opencv’, for both real-time video input as well as reference video. The body angles are calculated by taking the points for 3 joints, then finding the distance between them and finally computing the angle in between the joints. The joint points have been obtained using mediapipe.pose module and a custom function has been defined for calculating the angles. Our model works by frame-by-frame comparison of body angles. The body joint angles are calculated only for those joints which are visible in the reference video. Our model makes predictions in the form of ‘correct’ or ‘incorrect’. The body angles have been allowed to be correct at a deviation of 30 degrees, 20 degrees and 10 degrees to each side of the correct angle. The model has been evaluated by calculating the difference between the angles in the input frame and the reference frame for each joint, and then passing the dataframe for classification using supervised machine learning algorithms.

System/workflow architecture

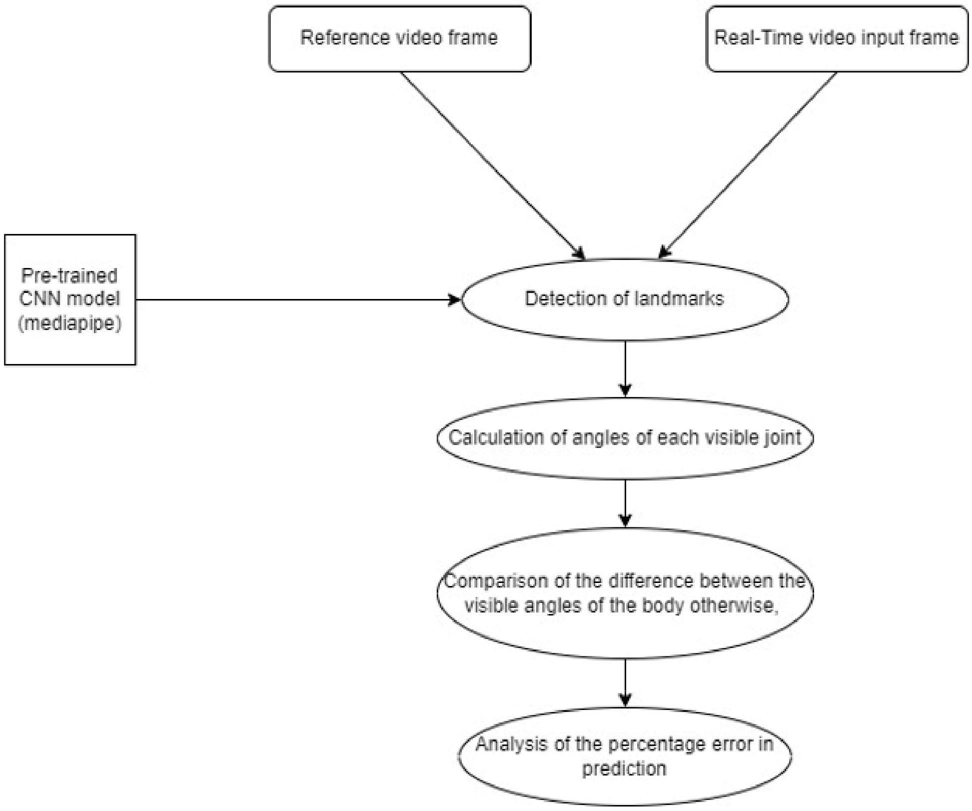


Fig. 2. Workflow diagram for our project

Pre-trained Convolutional Neural Network (CNN) model, specifically the MediaPipe model is used for landmark detection. Real time video of the person doing the yoga is captured through the webcam and reference video is embedded Mp4 file.This model can detect and track various landmarks, such as joints in the human body or facial features, in real-time input frames and reference video frames.Once the landmarks are detected both on the real time input and reference video input frames angles of all visible joints are calculated and stored in explicit variables.Difference of these angles are taken as criteria to label the person doing the yoga pose as correct or incorrect.When this angles difference falls within a certain range or deviation then label it as correct or else incorrect.The deviation in angles taken for labeling are 10,20,30 degrees.The percentage error is then calculated based on the number of incorrect joints compared to the total number of visible joints. This gives an idea of how well the input frame matches the reference frame in terms of joint angles.

<u>Performance Analysis</u>

Libraries Used:

cv2: ‘Cv2’ is OpenCV library which is mainly used when dealing with Image and Video I/O operations. In our project ‘cv2’ has been used for obtaining real-time video input from the web-cam, as well as for reading images (.jpg) and video (.mp4) files.

mediapipe: ‘mediapipe’ is a pre-trained machine learning model developed by Google that makes use of convolution neural networks (CNN) to detect and estimate the positions of key body landmarks or joints. These landmarks typically include the locations of the head, neck, shoulders, elbows, wrists, hips, knees, and ankles.

Opencv and mediapipe libraries are used to capture video from the default camera to detect human poses using the mediapipe pose estimation model and to render the pose landmarks and connection on the video feed.with videoCapture object is used to capture video from default camera. Mediapipepose Object with tracking confidence and minimum detection is set to 0.5 will be used to detect human poses in each frame of the video feed. After capturing the live video,convert the color space of the frame from BGR (which OpenCV uses by default) to RGB, which is the color space used by the Mediapipe library.To avoid modifications in the images mediapose detection is used.After conversion of image into RGB ,mp_drawing will help to render detections.

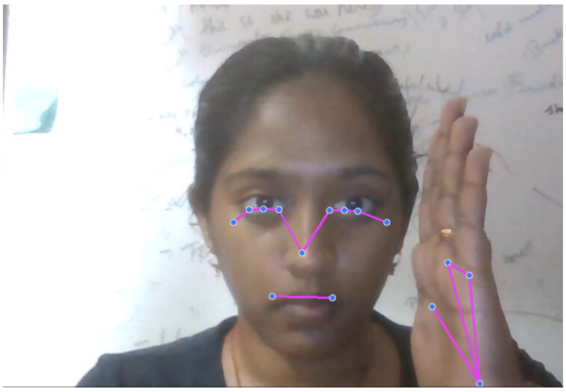
MediaPipe is a cross-platform framework for building multimodal applied ML pipelines. The PoseLandmark enum class is a set of 33 predefined human pose landmarks that can be detected by the MediaPipe Pose estimation module. These landmarks represent key points on the human body, such as the nose, eyes, ears, shoulders, elbows, wrists, hips, knees, and ankles.

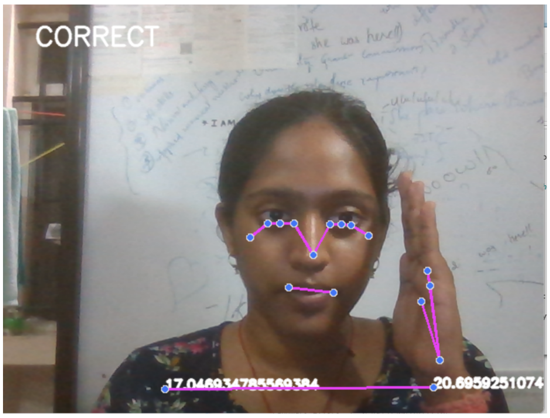
Calculating angles from the points from the video feed

This function calculates the angles between the points and then converts these coordinates into tuples. The arctan2 function is used to find the angles between the points. The angle is converted from radians into degrees.

Yoga pose image to image detection:

Using this mediapipe ,the landmarks are used to identify the key points of a person's body in an image or video. This can help you determine whether the person is performing a yoga pose correctly and can provide feedback for adjustments to improve their form.After live capturing of person doing yoga ,landmarks are eventually detected.X and y coordinates of Shoulder,elbow,wrist are extracted from the landmarks which are before detected while processing the image.calculate the angle using the shoulder,elbow,wrist X and y coordinates.





Live video frame to frame comparison and detection of Yoga pose correctly:

Yoga MP4 is already recorded and embedded in the local machine or jupyter notebook before executing the code.

Initially OpenCV and MediaPipe Pose are used to extract landmarks from webcam and Yoga MP4 video file.

VideoCapture object is used to capture video from webcam and another VideoCapture object is used to capture video from an yoga MP4 file.Script then sets sizes the window for the webcam and MP4video so that both width and height are equal.Both are resized to match webcam frame size and they are recolored to RGB.Images are processed using MediaPipe instance to detect the landmarksThe same process of processing is done for each frame of the MP4 video frame.The x and y coordinates of left and right

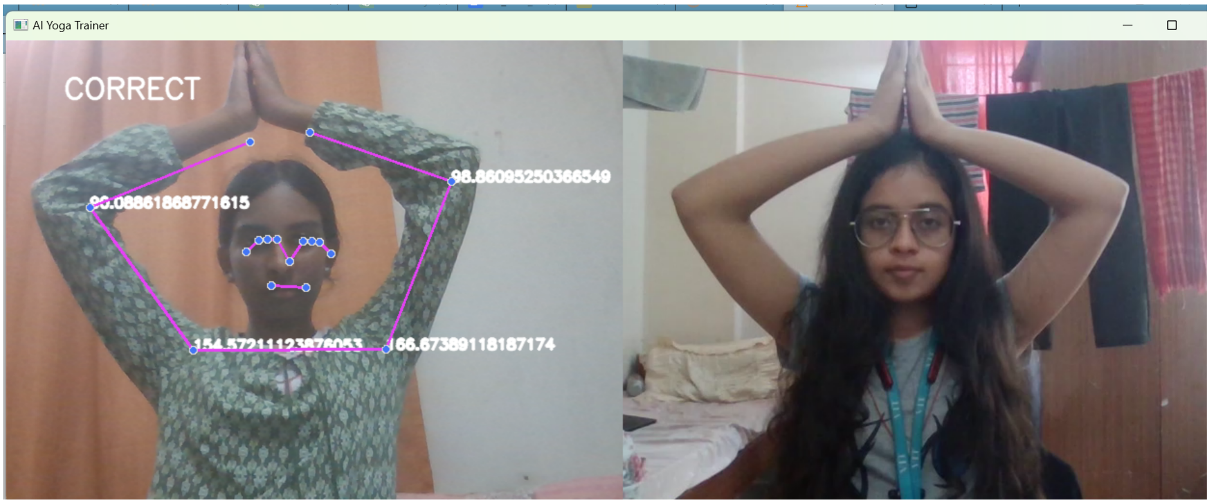
shoulder,elbow,wrist,hip,knee,ankle for both webcam and Yoga video frames are extracted from landmarks.After rendering detections all frames set back to BGR frame.After extracting the coordinates of all the landmarks present in the frames like Shoulder,hip,elbow,wrist,knee,ankle for the frames of the live capturing of the person who is doing yoga,extract the coordinates of the landmarks in reference video frames also.
Calculate angles for all the landmarks using the coordinates extracted for the video capturing of the person doing yoga.

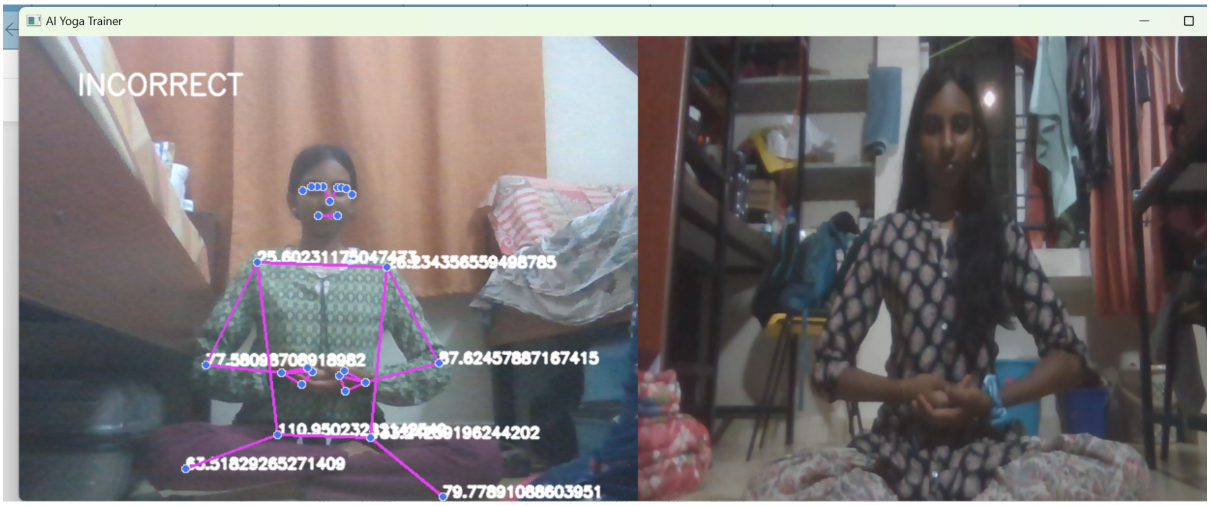
left_elbow_angle = calculate_angle(left_shoulder, left_elbow, left_wrist)

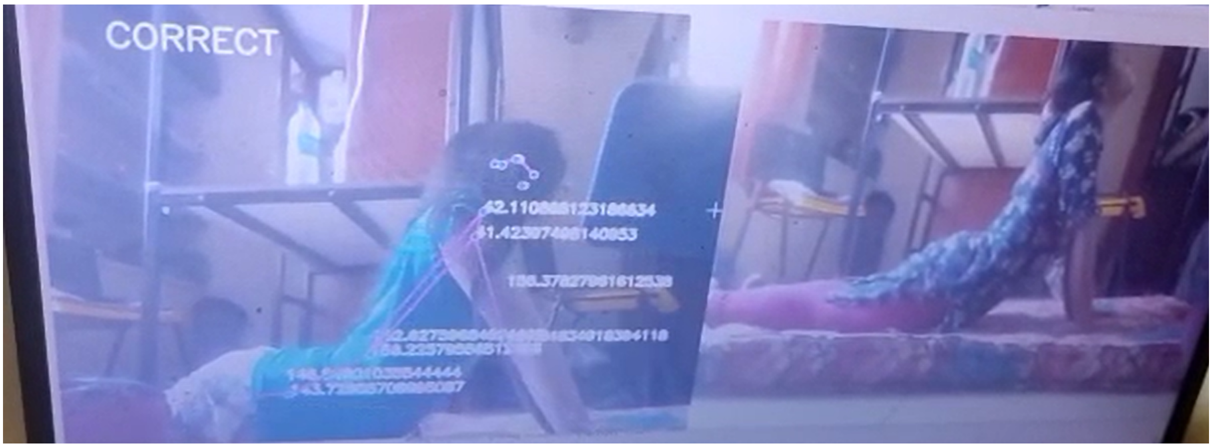
Here in the live capturing elbow and shoulder are detected so formulae for calculating the elbow angle for the person.Iterate this process for all the landmarks detected in the live capturing of a person doing yoga using webcam.Do the same task for all the frames in the reference video.(yoga video)

ileft_elbow_angle = calculate_angle(ileft_shoulder, ileft_elbow, ileft_wrist)

At an instant shoulder,elbow,wrist are landmarks detected in single frame of the reference video,left elbow angle is calculated using the coordinates extracted from the landmarks of the reference frame.After calculating all the angles visualize them so that the detections appear on the screen.







Evaluation of the yoga pose by user according to the reference video:

The angles present in each frame of the real-time video input are calculated and compared with each corresponding frame of the reference video. The angles present in each frame of the real-time video input are matched with the corresponding frame of reference video with deviations upto 10 degrees, 20 degrees and 30 degrees. At each run, the difference in the corresponding joint angles of the input video frames and reference video frames are calculated and

stored in a dataframe. This dataframe is then used to calculate the accuracy of the model. A 'prediction' column is appended to the dataframe, where 0 indicates 'Incorrect' and 1 indicates 'correct'.
The whole dataset is validated using 5-fold cross-validation in which a few machine learning algorithms, namely, Logistic Regression, Support Vector Machine Classifier, K-Nearest Neighbours Classifier, Decision Tree Classifier, Multi-Layer Perceptron Classifier and Random Forest Classifier.
However, our model is a learning model and keeps improving itself everytime it is run. The records for each run keep getting stored in the dataframe and as such a huge amount of data is available for evaluating and improving the accuracy.

Model Accuracy

The angles inputted from each yoga pose is stored in a list separately. These lists are combined together in a dataframe . The dataframe consists of:

0 LeftElbowAngle 460 non-null float64
1 RightElbowAngle 460 non-null float64
2 LeftShoulderAngle 460 non-null float64
3 RightShoulderAngle 460 non-null float64
4 LeftHipAngle 460 non-null float64
5 RightHipAngle 460 non-null float64
6 LeftKneeAngle 460 non-null float64
7 RightKneeAngle 460 non-null float64
8 Prediction 460 non-null float64

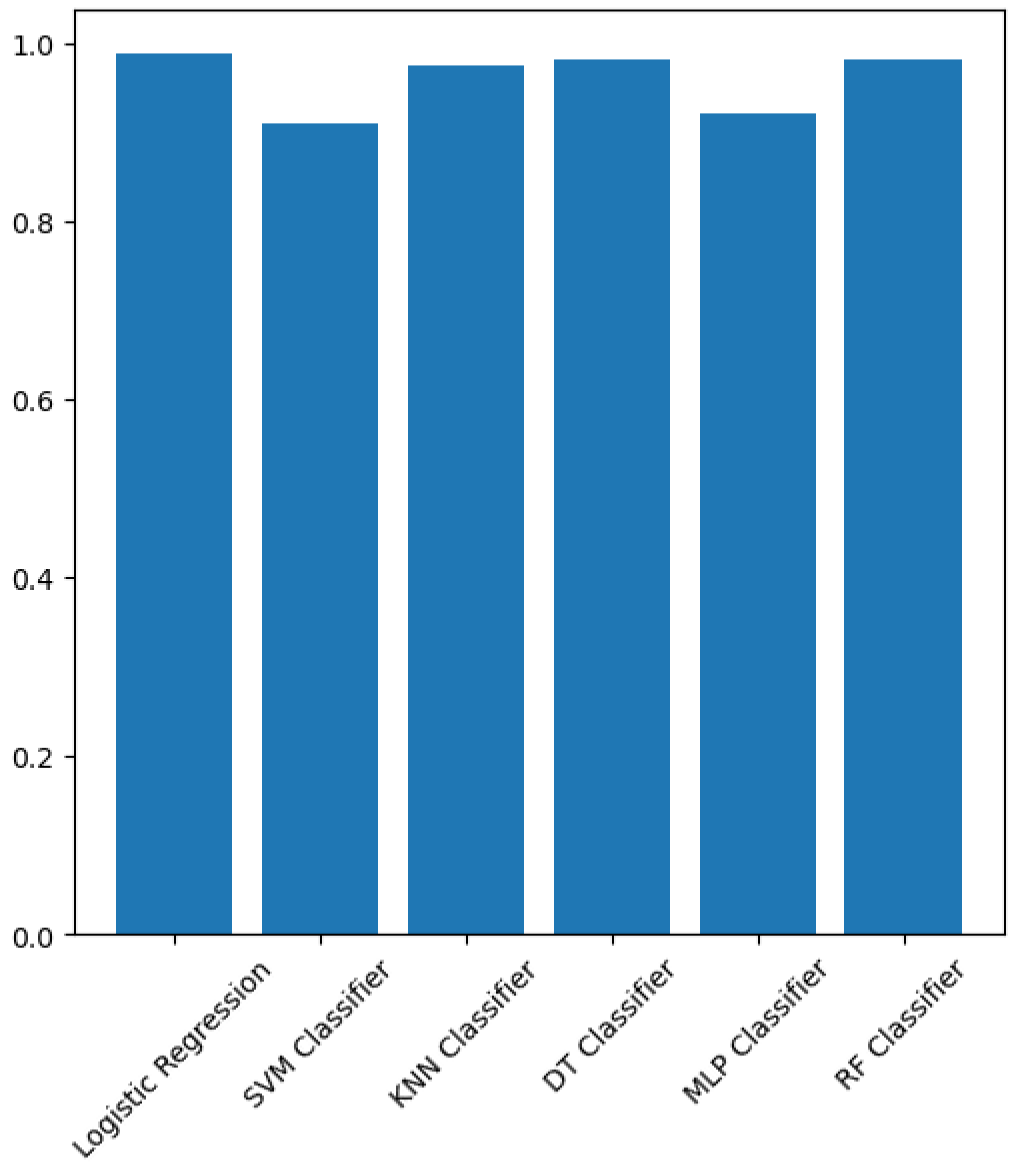
Where each column is to datatype of float64.The dataframe is then checked for null values. The rows containing the null values are dropped.

ML algorithm	±10° dev	±20° dev	±30°dev
Logistic Regression	0.99	0.94	0.99
SVM Classifier	0.90	0.94	0.99
K Nearest Neighbours Classifier	0.98	0.95	0.99
Decision Tree Classifier	0.98	0.98	0.98

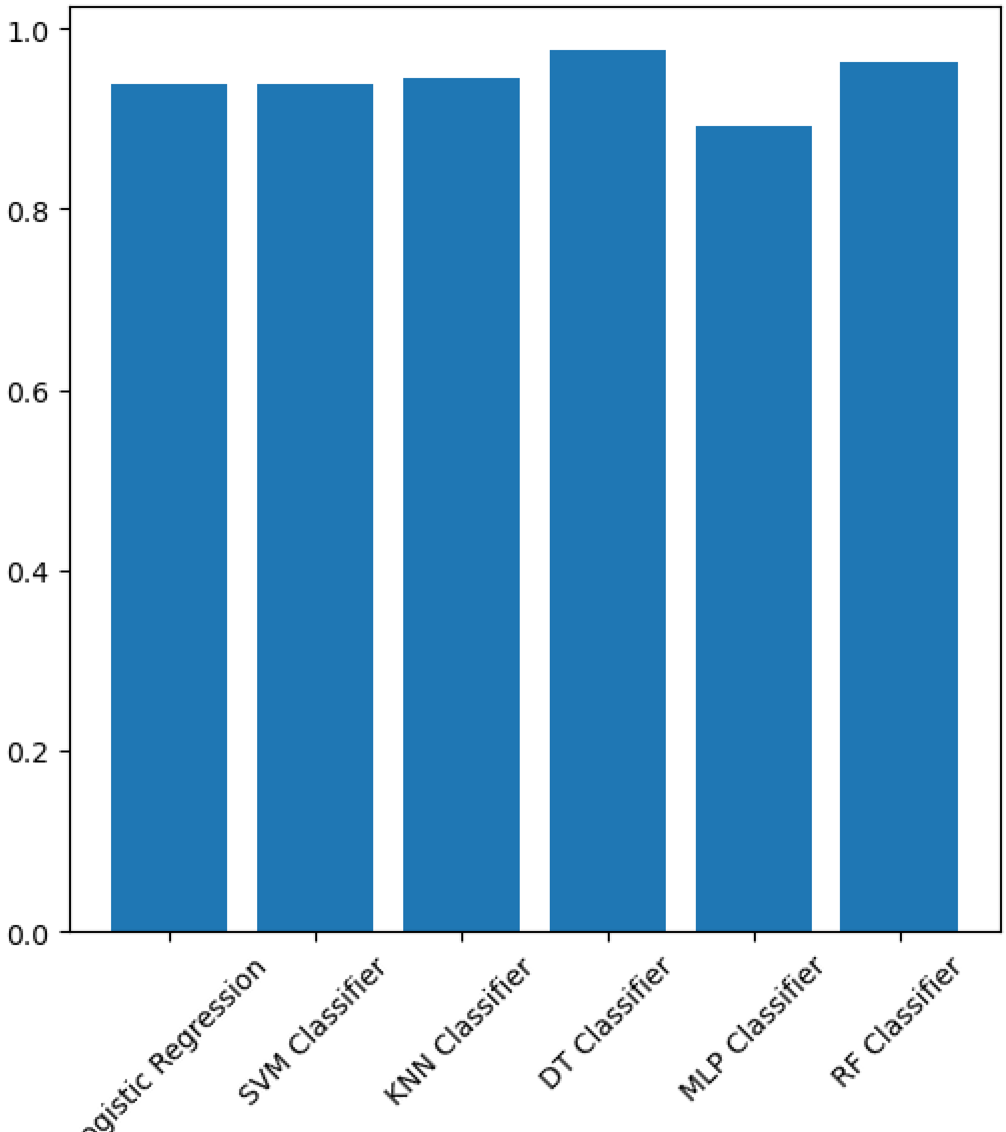
MLP Classifier	0.92	0.89	0.99
Random Forest Classifier	0.98	0.96	0.99

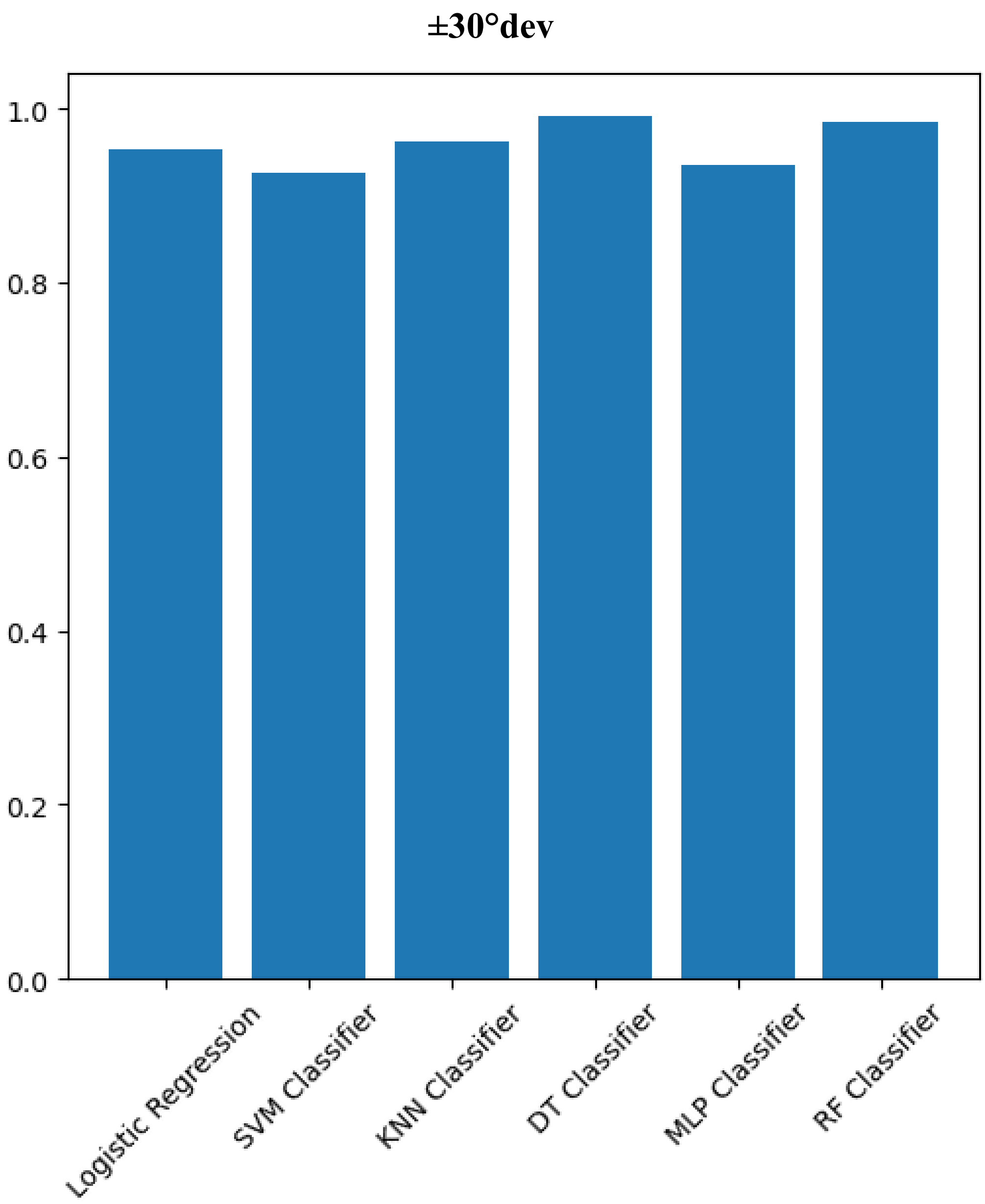
Below are the visualisation for each:

±10° dev



±20° dev


1.0
0.8
0.6
0.4
0.2
0.0
Logistic Regression
SVM Classifier
KNN Classifier
DT Classifier
MLP Classifier
RF Classifier


±30°dev
1.0
0.8
0.6
0.4
0.2
0.0
Logistic Regression
SVM Classifier
KNN Classifier
DT Classifier
MLP Classifier
RF Classifier

Conclusion

We have successfully implemented an AI yoga trainer by incorporating comparison of real-time video input with reference yoga video. As we know, that most of the research articles that have been published have just applied classification models on a limited dataset consisting of yoga pose pictures for classifying the yoga poses performed by the users, we have achieved novelty by allowing frame-to-frame comparison of real-time input video with reference video.

In conclusion, incorporating AI technology into yoga training can offer numerous benefits to both yoga trainers and practitioners. AI can assist in personalizing yoga sessions based on the practitioner's unique needs, abilities, and goals, thus enhancing the effectiveness of the training. Additionally, AI can provide real-time feedback and corrections to the practitioner, reducing the risk of injury and improving their overall performance. Moreover,this will help create customized training sessions for the people. Therefore, while AI can support yoga training, it should not replace the critical role of the yoga trainer in providing guidance, motivation, and emotional support to the practitioner.

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